

EC466/966: Estimation and Inference in Econometrics

Exercise # 3- Answer Key

1. Please check EC966_466hw3_sol.py for the answers.
2. Suppose that you have the following regression model

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where $\mathbf{y}$ is the $n \times 1$ vector of the dependent variable, $\mathbf{X}$ is the $n \times K$ vector of regressors, β is the $K \times 1$ vector of the model parameters, and ϵ is the $n \times 1$ vector of the error term. Assume that Classical Linear Regression Model assumptions hold except the homoskedasticity assumption, i.e., $E(\epsilon\epsilon'|\mathbf{X}) = \sigma^2\mathbf{V}$ where $\mathbf{V}$ is a symmetric and positive definite matrix, and is a function of $\mathbf{X}$.

- (a) Derive the Generalized Least Squares (GLS) estimator. [a] The GLS estimator is obtained by the below minimization problem

$$\min_{\tilde{\beta}} (\mathbf{y} - \mathbf{X}\tilde{\beta})'(\sigma^2\mathbf{V})^{-1}(\mathbf{y} - \mathbf{X}\tilde{\beta}) = \min_{\tilde{\beta}} WSSR(\tilde{\beta}).$$

The first order condition is

$$\frac{\partial WSSR(\tilde{\beta})}{\partial \tilde{\beta}} = -2\mathbf{X}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}_{GLS}) = \mathbf{0}.$$

Then

$$\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\hat{\beta}_{GLS} = \mathbf{X}'\mathbf{V}^{-1}\mathbf{y}.$$

$\mathbf{X}$ is a full column rank matrix and $\mathbf{V}^{-1}$ is a positive definite matrix; therefore, $\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}$ is a positive definite matrix and invertible. Thus, we have

$$\hat{\beta}_{GLS} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} \mathbf{X}'\mathbf{V}^{-1}\mathbf{y}.$$

- (b) Show that the GLS is an (conditionally) unbiased estimator. [b] Note

that $\mathbf{V}$ is function of $\mathbf{X}$. Therefore,

$$\begin{aligned} E(\hat{\beta}_{GLS}|\mathbf{X}) &= E((\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}|\mathbf{X}) \\ &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}E(\mathbf{y}|\mathbf{X}) \\ &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\beta \\ &= \beta. \end{aligned}$$

- (c) Derive the variance-covariance matrix for the GLS estimator. [c] The variance-covariance matrix for the GLS estimator is

$$\begin{aligned} Var(\hat{\beta}_{GLS}|\mathbf{X}) &= Var((\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}|\mathbf{X}) \\ &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}Var(\mathbf{y}|\mathbf{X})\mathbf{V}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{V}\mathbf{V}^{-1}\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}. \end{aligned}$$

- (d) Is the OLS estimator unbiased in the presence of heteroskedasticity? [d] The OLS estimator is still unbiased, because

$$\begin{aligned} E(\hat{\beta}_{OLS}|\mathbf{X}) &= E((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}|\mathbf{X}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{y}|\mathbf{X}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta \\ &= \beta. \end{aligned}$$